

MACHINE LEARNING FUNDAMENTALS - LS2026

SEMINAR: INTRODUCTION

CZECH TECHNICAL UNIVERSITY IN PRAGUE
V. FRANC

Assignment 1. Consider practical machine learning applications that require learning predictors from data. Provide examples of real-world scenarios in which the development (training/validation) data and the unseen (deployment) data are:

- a) Independent and identically distributed (i.i.d.)
 - b) Not independent and/or not identically distributed (non-i.i.d.)
- For each case, briefly explain why the i.i.d. assumption holds or does not hold.

Solution 1. a) Development and deployment data are i.i.d.

Example 1: Handwritten Digit Recognition.

- Task: Classify images of handwritten digits.
- Training data: Randomly sampled scanned digit images.
- Deployment data: New scanned digit images from the same source.
- Why i.i.d.?
 - Each image is sampled independently.
 - The distribution of handwriting styles and preprocessing remains the same.
 - Training and deployment data come from the same population.

Example 2: Email Spam Classification (Single Organization).

- Task: Predict whether an email is spam.
- Training data: Historical emails from employees.
- Deployment data: Future incoming emails within the same organization (short-term horizon).
- Why i.i.d.?
 - Emails are treated as independent samples.
 - The statistical properties of emails remain stable.
 - No systematic distribution shift occurs.

Example 3: Industrial Visual Inspection.

- Task: Detect defects in manufactured parts.
- Training data: Images collected from a production line.
- Deployment data: Future images from the same production line under identical conditions.
- Why i.i.d.?
 - Same camera, lighting, materials, and process.
 - Independent sampling of parts.
 - Stable data-generating distribution.

b) Development and deployment data are not i.i.d.

This occurs when:

- Observations are dependent (e.g., temporal or spatial correlation).
- The training and deployment populations differ (distribution shift).

Example 1: Autonomous Driving.

- Training data: Collected in sunny conditions.
- Deployment data: Used in snowy or rainy conditions.
- Why not i.i.d.?
 - Weather and lighting conditions differ.
 - The input distribution changes (covariate shift).

Example 2: Medical Diagnosis Across Hospitals.

- Training data: Data from one hospital.
- Deployment data: Data from another hospital with different demographics and equipment.
- Why not i.i.d.?
 - Patient populations differ.
 - Measurement processes differ.
 - Disease prevalence may differ (label shift).

Example 3: Stock Price Prediction.

- Training data: Historical time series.
- Deployment data: Future time points.
- Why not i.i.d.?
 - Observations are temporally correlated (not independent).
 - Market conditions evolve over time.

Example 4: Recommendation Systems.

- Training data: Past user interactions.
- Deployment data: Future interactions influenced by the deployed model.
- Why not i.i.d.?
 - User behavior changes over time.
 - The model influences future data (feedback loop).
 - Samples are not independent.

Assignment 2. Consider the task of age estimation from a facial image. Let the face image be denoted by $x \in \mathcal{X}$ and the (unknown) age by $y \in \mathcal{Y}$. Assume that the statistical relationship between these two random variables is known and described by their joint distribution $p(x, y)$. Let $h : \mathcal{X} \rightarrow \mathcal{Y}$ be a predictor.

Interpret the true risk

$$R(h, p) = \mathbb{E}_{(x, y) \sim p(x, y)} [\ell(y, h(x))],$$

and derive the optimal Bayes predictor (i.e., the predictor that minimizes $R(h, p)$) for the following settings:

a) $\mathcal{Y} = \{0, 1, \dots, Y\}$ and the 0/1 loss

$$\ell(y, y') = \mathbb{I}[y \neq y'].$$

b) $\mathcal{Y} = \{0, 1, \dots, Y\}$ and the loss

$$\ell(y, y') = \mathbf{1}\{|y - y'| > 5\}.$$

c) $\mathcal{Y} = [0, 100]$ and the squared loss

$$\ell(y, y') = (y - y')^2.$$

d) $\mathcal{Y} = [0, 100]$ and the absolute deviation loss

$$\ell(y, y') = |y - y'|.$$

Solution 2. Solution.

The true risk:

$$R(h, p) = \mathbb{E}_{(x, y) \sim p}[\ell(y, h(x))] = \int_{\mathcal{X}} p(x) \sum_y p(y | x) \ell(y, h(x)) dx$$

The Bayes predictor:

$$h^*(x) = \arg \min_{\hat{y}} \sum_y p(y | x) \ell(y, \hat{y}) = \arg \min_{\hat{y}} r_x(\hat{y})$$

a)

$$\ell(y, \hat{y}) = \mathbb{I}[y \neq \hat{y}]$$

$$r_x(\hat{y}) = \sum_y p(y | x) \mathbb{I}[y \neq \hat{y}] = 1 - p(\hat{y} | x)$$

$$\hat{y}_* = \arg \max_{\hat{y}} p(\hat{y} | x)$$

b)

$$\ell(y, \hat{y}) = \mathbb{I}[|y - \hat{y}| > 5]$$

$$r_x(\hat{y}) = \sum_y p(y | x) \mathbb{I}[|y - \hat{y}| > 5] = 1 - \mathbb{P}(|Y - \hat{y}| \leq 5 | X = x)$$

$$\hat{y}_* = \arg \max_{\hat{y}} \sum_{y: |y - \hat{y}| \leq 5} p(y | x)$$

c)

$$\ell(y, \hat{y}) = (y - \hat{y})^2$$

$$r_x(\hat{y}) = \int p(y | x) (y - \hat{y})^2 dy$$

$$= \int p(y | x) (y^2 - 2\hat{y}y + \hat{y}^2) dy$$

$$\begin{aligned}
&= \int p(y | x) y^2 dy - 2\hat{y} \int p(y | x) y dy + \hat{y}^2 \\
&\frac{\partial r_x(\hat{y})}{\partial \hat{y}} = -2 \int p(y | x) y dy + 2\hat{y} = 0 \\
&\hat{y}_* = \int p(y | x) y dy = \mathbb{E}[y | x]
\end{aligned}$$

d)

$$\begin{aligned}
\ell(y, \hat{y}) &= |\hat{y} - y| \\
r_x(\hat{y}) &= \int p(y | x) |\hat{y} - y| dy \\
\hat{y}_* &= \arg \min_{\hat{y}} r_x(\hat{y}) = \text{Median}(y | x) = M_x \\
\mathbb{P}(y \leq x | x) &\geq \frac{1}{2} \quad \text{and} \quad \mathbb{P}(y \geq x | x) \geq \frac{1}{2}
\end{aligned}$$

Assignment 3. Consider a binary classification problem with scalar observation $\mathcal{X} = \mathbb{R}$, two possible classes $\mathcal{Y} = \{-1, +1\}$ and the 0/1-loss $\ell(y, y') = \mathbb{I}[y \neq y']$. The observations for both classes are generated according to Gaussian distributions. Specifically, the joint probability distribution of the observation $x \in \mathbb{R}$ and the class label $y \in \mathcal{Y}$ is given by:

$$p(x, y) = p(y) \frac{1}{\sqrt{2\pi}\sigma_y} \exp\left(-\frac{1}{2\sigma_y^2}(x - \mu_y)^2\right), \quad y \in \mathcal{Y},$$

where $p(y)$ is the prior distribution of the class y , μ_+ and μ_- are the means of the distributions for $y = +1$ and $y = -1$, respectively, $\sigma_+ > 0$ and $\sigma_- > 0$ are the corresponding standard deviations.

a) Assume $\mu_- < \mu_+$ and $\sigma_+ = \sigma_-$. Show that, under these conditions, the Bayes optimal prediction strategy is a thresholding rule of the form:

$$h(x) = \begin{cases} -1 & \text{if } x < \theta, \\ +1 & \text{if } x \geq \theta, \end{cases}$$

where $\theta \in \mathbb{R}$ is a scalar threshold. Derive explicit formula for computing θ .

b) Now assume $\mu_+ = \mu_-$ and $\sigma_+ \neq \sigma_-$. Determine the optimal prediction strategy under these conditions.

c*) Finally, consider the case where $\mu_+ = \mu_-$, $\sigma_+ \neq \sigma_-$, and both classes have nonzero prior probabilities, i.e. $p(+1) > 0$ and $p(-1) > 0$. Is it possible for the Bayes classifier to assign all inputs $x \in \mathbb{R}$ to a single class? Prove your answer.

Solution 3. The Bayes classifier in case of the 0/1-loss and two classes assigns the input x into the class with the higher class posterior $p(y | x)$, or equivalently with the higher $p(x, y)$, that is,

$$h^*(x) = \begin{cases} +1 & \text{if } p(x, y = +1) > p(x, y = -1) \\ -1 & \text{if } p(x, y = +1) < p(x, y = -1) \end{cases}$$

Note that the boundary inputs, $p(x, y = +1) = p(x, y = -1)$, can be assigned to an arbitrary class. Let us define a discriminant function $f(x)$ as a logarithm of the likelihood ratio:

$$f(x) = \log \left(\frac{p(x, y = +1)}{p(x, y = -1)} \right) . \quad (1)$$

The Bayes classifier can be expressed equivalently as the sign of the discriminant function:

$$h^*(x) = \text{sign}(f(x)) .$$

After substituting

$$p(x, y) = p(y) \frac{1}{\sqrt{2\pi}\sigma_y} \exp \left(-\frac{1}{2\sigma_y^2} (x - \mu_y)^2 \right)$$

to (1) we get

$$\begin{aligned} f(x) &= \log p(x, y = +1) - \log p(x, y = -1) \\ &= \log \frac{p(+1)}{\sigma_+} - \log \sqrt{2\pi} - \frac{(x - \mu_+)^2}{2\sigma_+^2} - \log \frac{p(-1)}{\sigma_-} + \log \sqrt{2\pi} + \frac{(x - \mu_-)^2}{2\sigma_-^2} \\ &= x^2 \underbrace{\left(\frac{1}{2\sigma_-^2} - \frac{1}{2\sigma_+^2} \right)}_a + x \underbrace{\left(\frac{\mu_+}{\sigma_+^2} - \frac{\mu_-}{\sigma_-^2} \right)}_b + \underbrace{\frac{\mu_-^2}{2\sigma_-^2} - \frac{\mu_+^2}{2\sigma_+^2} + \log \frac{p(+1)\sigma_-}{p(-1)\sigma_+}}_c \\ &= a x^2 + b x + c . \end{aligned}$$

a) In case of $\sigma_+ = \sigma_- = \sigma$, the multiplier in front of x^2 is $a = 0$ and the discriminant function becomes a linear function. Hence,

$$h^*(x) = \text{sign}(f(x)) = \text{sign}(b x + c) = \begin{cases} -1 & \text{if } x < \theta \\ +1 & \text{if } x \geq \theta \end{cases}$$

where θ is a solution of the linear equation $f(x) = b x + c = 0$, i.e.

$$\theta = -\frac{c}{b} = \frac{\mu_+ + \mu_-}{2} + \frac{\sigma^2}{\mu_- - \mu_+} \log \frac{p(+1)}{p(-1)} .$$

b) In case of $\mu_+ = \mu_- = \mu$ we can rewrite the discriminant function as

$$f(x) = \frac{1}{2} \left(\frac{1}{\sigma_-^2} - \frac{1}{\sigma_+^2} \right) (x - \mu)^2 + \log \frac{p(+1)\sigma_-}{p(-1)\sigma_+} .$$

If $\sigma_- < \sigma_+$, the discriminant function is convex. If $p(+1)\sigma_- < p(-1)\sigma_+$, then the minimum of $f(x)$ is a negative number and the quadratic equation $f(x) = 0$ has two solutions, which we denote θ_1 and θ_2 . In this case, the Bayes classifier assigns the inputs x that fall to the interval $[\theta_1, \theta_2]$ into the negative class and the inputs outside the interval to the positive class. If $p(+1)\sigma_- > p(-1)\sigma_+$, the minimum of $f(x)$ is a positive number and the Bayes classifier assigns all inputs to the negative class. If $\sigma_- > \sigma_+$, the discriminant function is concave and the analysis is analogous.

c*) Yes, it can happen. For example, when $\mu_+ = \mu_-$, $\sigma_- < \sigma_+$ and $p(+1)\sigma_- > p(-1)\sigma_+$, the discriminant function attains a positive values for all x and hence all inputs are assigned to the positive class.